A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 31

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (52 - 40) / 40 x 100 **1** = 30% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Sulfur dioxide came mainly from burning coal – domestic coal fires were controlled by the Clean Air Acts and coal-fired power stations have closed or been fitted with flue gas desulfurisation **1** Low-sulfur fuels are now used, and sulfur can be removed from the fuel before combustion **1** Nitrogen oxides are produced whenever any fuel is burned in air at high temperature, so they are not removed by changing the fuel – road traffic in the city centre has continued to grow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **flue gas desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone	3
01.3	**1 mark each** 1 mark for each of the 3 points below. **1 Electrostatic precipitator:** the flue gases pass charged wires that give the particulates an electric charge, so they are attracted to oppositely charged collecting plates and can be shaken off and removed **1 Catalytic converter:** a platinum / rhodium catalyst converts oxides of nitrogen, carbon monoxide and unburnt hydrocarbons into nitrogen, carbon dioxide and water **1 Low-NOx burner:** the fuel is burned at a lower temperature, or in stages, so less of the nitrogen and oxygen in the air combines to form oxides of nitrogen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Electrostatic precipitator** → charged plates that attract and collect particulates • **particulates** → fine solid particles / smoke and dust • **Catalytic converter** → a catalyst that converts NOx, CO and unburnt hydrocarbons to nitrogen, carbon dioxide and water	3
01.4	**1 mark each** 1 mark for each of the 3 points below. **1** Ground-level ozone is a secondary pollutant: it is not emitted directly but formed in the air by sunlight acting on oxides of nitrogen and volatile organic compounds **1** Nitrogen dioxide is split by ultraviolet light to give nitrogen monoxide and an oxygen atom, which joins an oxygen molecule to make ozone **1** With less NOx there is less of the raw material for these photochemical reactions, so less ozone and fewer PANs are produced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • ground-level ozone described as protecting us from UV - that is stratospheric ozone; at ground level it is a pollutant	3

Do not write outside the box

Question	Marking guidance	Marks
01.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Lichens differ in their sensitivity to sulfur dioxide, so the species present show the average pollution level over a long period rather than at one instant **1** They need no power supply or expensive equipment and can be surveyed at many sites, giving good spatial coverage cheaply **1** They integrate the effects of pollution over years, so a short pollution episode may not show up **1** They indicate sulfur dioxide well but say little about nitrogen dioxide, ozone or particulates, which are now the city's main problems **1** Other factors affect them too, such as the pH and texture of the bark or stone they grow on, shade and humidity, so a survey can be misleading unless these are standardised **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **indicator species** → a species whose presence or absence shows the conditions present • **particulates** → fine solid particles / smoke and dust **Do not credit** • 'lichens die where the air is polluted' without naming the pollutant - it is their sensitivity to sulfur dioxide that is used	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-254; AQA spec 3.4.3.2.1-3.4.3.2.5.	
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - tiny pores between clay particles allow capillary rise – none in sand	1
02.2	**1 mark each** 1 mark for each of the 3 points below. **1 Hygroscopic:** held tightly by adhesive forces on particle surfaces – cannot drain and cannot be absorbed by plants **1 Capillary:** held in pores by surface tension – does not drain but can be absorbed by plants **1 Gravitational:** drains freely through large pores – available to plants only briefly as it passes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.3	1 mark for the point below. **1** Small round crumbs leave spaces giving good drainage, aeration and easy root penetration – large flat platy peds pack together reducing all three **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **peds** → soil crumbs / aggregates of soil particles	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Tropospheric ozone (formed by reactions of NOx in sunlight)	1
03.2	**1 mark each** 1 mark for each of the 4 points below. **1 Primary pollutant:** released directly by human activities, eg sulfur dioxide from burning coal / smelting sulfide ores – NOx from high-temperature combustion **1 Secondary pollutant:** produced by chemical reactions between primary pollutants, often with non-pollutants (water, oxygen, sunlight) **1** Sulfur dioxide is oxidised by ozone to sulfur trioxide which dissolves to form sulfuric acid – NOx dissolve to form nitric acid **1** The acids (sulfurous, sulfuric, nitric) are secondary pollutants – ozone itself is a secondary pollutant of NOx photochemistry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 237-238; AQA spec 3.4.1.	
04.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 B:** Oxygen	1
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** A change to one process causes other processes to alter as a direct result (chains of linked processes / feedback) **1** There is still a lot that is not understood about how atmospheric processes work **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** The surface warmed in the tropics heats the air above – warm winds (south-westerlies from the Caribbean) carry heat energy to higher latitudes **1** Winds carry water vapour (evaporated from warm oceans) to areas that would otherwise receive little precipitation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-88; AQA spec 3.2.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 6 points below. **1 Location:** discharge into the sea / large water bodies where currents dilute and disperse the pollutant **1 Location:** chimneys / emissions sited downwind of urban areas so pollutants are not carried over populations – avoiding valleys and enclosed water bodies where pollutants concentrate **1 Location:** not discharging wastes onto permeable rock above an aquifer / not siting landfill on aquifers **1 Timing:** restrict polluting activities (industrial combustion, vehicle use) during temperature inversions when dispersal is poor **1 Timing:** discharge liquid effluent when river flow is high / on an ebb tide so it is diluted and carried away **1 Timing:** pesticide spraying at night / on still days – noisy operations restricted to daytime – night-flight restrictions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **permeable** → how easily water passes through / lets water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	6
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** The organisation / person that causes the pollution is responsible for (paying for) the damage caused **1** If the cost of preventing the pollution is less than the cost of the damage / fines, there is a financial incentive to prevent it – works best when penalties are high and escape is unlikely **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** It is assumed that the waste will cause pollution / harm until research shows it is unlikely to **1** So release is not permitted until it has been tested – lack of knowledge of a problem is not an excuse / does not reduce responsibility **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 4 points below. **1 Wheel vibration:** track polishing / grinding, sound-absorbing ballast, sound-absorbing suspension **1 Pantograph:** aerodynamic fairing **1 Wheel squeal:** lubrication of wheels / track on corners **1 Braking:** composite-material brakes instead of cast iron **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
06.2	1 mark for the point below. **1** Roads pass through / near almost all residential areas and traffic is continuous through the day, unlike airports or railways which affect limited areas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
06.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Sound-absorbing road surfaces (porous asphalt / rubber-crumb asphalt) **1** Acoustic fences, embankments / baffle mounds, walls or dense tree planting **1** Traffic management for free flow (less acceleration noise) – re-routing traffic away from residential areas **1** Double glazing / acoustic insulation of buildings **1 Vehicle design:** better aerodynamics, engine insulation, quieter exhausts, quieter tyres **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Air bubble curtains around the pile driver absorb noise **1** Acoustic survey to detect cetaceans first and delay work if present – increase volume gradually (soft start) so animals can leave **1** Drilling instead of pile-driving **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 292-294; AQA spec 3.4.3.2.13.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** Small seeds, genetically uniform, high and rapid germination rate so results are quick and reliable **1** Wild species have different ranges of tolerance so cress results cannot be applied directly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated • **tolerance** → the range of conditions in which a species can survive **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Preliminary study with widely spaced pH values to find where germination occurs **1** Final range from below the lowest pH giving germination to above the highest, with closer intervals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.3	**1 mark each** 1 mark for each of the 4 points below. **1** Temperature (constant, natural germination temperature) **1** Light, water volume, number of seeds per dish, seed source **1** Growth medium (filter paper; soil would alter pH) – nutrients not needed for germination **1** Continue until it is clear that ungerminated seeds will not germinate, so that a delay in germination is not mistaken for failure **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Surface water affects fresh mass and drying damages the plants – dry mass requires killing the plants so the same plants cannot be re-measured **1** Leaf area is a reasonable estimate of biomass that allows repeated measurement of the same plants over time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 416-418; AQA spec 3.7.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 2 points below. **1 Negative feedback:** a change causes other changes that reduce the rate or size of the original change, re-establishing equilibrium **1 Positive feedback:** a change causes other changes that increase the original change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **Negative feedback** → a change that cancels itself out / a self-correcting cycle	2
08.2	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Increased low-level cloud:** higher temperatures increase evaporation and condensation, producing more cloud – clouds have a high albedo so more sunlight is reflected and warming is reduced **2 Increased photosynthesis:** higher temperatures and CO2 increase photosynthesis, removing CO2 – if stored in woody tissue atmospheric CO2 rises less **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **negative feedback** → a change that cancels itself out / a self-correcting cycle • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	4
08.3	**1 mark each** 1 mark for each of the 4 points below. **1 Permafrost:** frozen waterlogged Arctic soils contain dead organic matter that decomposed anaerobically producing methane trapped in the frozen ground **1** Warming thaws the permafrost, releasing methane (a powerful greenhouse gas) and allowing further decomposition, causing more warming **1 Methane hydrate:** methane from decomposed organic matter in deep sea sediments is held as a solid under high pressure and low temperature **1** If deep water warms the hydrate melts, releasing methane into the atmosphere and causing further warming **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **waterlogged** → soil pores filled with water so there is no air	4

Do not write outside the box

Question	Marking guidance	Marks
08.4	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Reduced albedo as snow / ice melts **1** Ocean acidification reducing coral survival and carbon sequestration **1** Increased forest and peat fires releasing CO2 **1** Faster decomposition of soil organic matter releasing CO2 **1** Increased water vapour (a greenhouse gas) in warmer air **1** Increased cirrus (high) cloud which traps infrared **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	3
08.5	**1 mark each** 1 mark for each of the 2 points below. **1** A point at which natural processes triggered by human-caused warming become self-sustaining / unstoppable **1** So that stopping the original human activity would no longer stop climate change (eg runaway permafrost melt, methane hydrate release, forest fires) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-104, 371-372; AQA spec 3.2.1.2, 3.6.1.	
09.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Disposal of items containing mercury:** batteries, thermometers, fluorescent lights **1** Chemical plants producing chlorine using mercury electrodes **1** Combustion of coal **1** Gold mining (amalgamation) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3

Do not write outside the box

Question	Marking guidance	Marks
09.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Replacement of mercury thermometers with alcohol / electronic thermometers (discontinued uses) **1** Removal from effluents by reverse osmosis / activated carbon filters / ion exchange filters **1** Disposal / storage of wastes at high pH to reduce solubility **1** Encapsulation of mercury-containing hazardous waste in cement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Most heavy metals are more soluble at low pH / less soluble at high pH **1** So at high pH they remain insoluble / immobile and cannot be leached into water or absorbed by organisms **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** The effects of two pollutants interact to produce a different (greater) effect than either alone **1** Cadmium and zinc – ozone and sulfur dioxide **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **synergism** → two pollutants together doing more damage than the sum of each alone **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-278; AQA spec 3.4.3.2.11.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Fossil fuel technologies:** secondary / tertiary recovery raising recovery from 20% to 60% – directional drilling and subsea wells – ROVs – fracking for tight oil and gas – tar sands / oil shales – coal gasification / liquefaction – enhanced gas recovery – methane hydrates – CCS (pre / post combustion, storage) • **Limits:** still finite (discoveries declining since the 1980s) – rising energy and financial cost of unconventional sources – environmental impacts (aquifers, water use, tremors, land take, habitat damage) – CO2 and other pollution – CCS unproven at scale and impractical for vehicles • **Nuclear:** new uranium extraction (polymer adsorption from seawater, phosphate mining, coal ash) – molten salt reactors – plutonium breeder and thorium reactors increasing fuel use many-fold – fusion as a long-term prospect • **Limits:** cost overruns, decommissioning, waste, public opposition, proliferation, embodied energy of low-grade uranium – fusion not commercial • **Sustainability criteria:** maintaining supply without unacceptable environmental, economic or social impact – comparison with renewables and energy conservation – energy gap argument • **Judgement:** technologies can extend supplies and reduce some impacts but cannot make non-renewables sustainable indefinitely	9

Do not write outside the box

Question	Marking guidance	Marks
	– role as transition / base-load alongside renewables **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **fracking** → pumping fluid into rock at high pressure to release gas or oil • **CCS** → capturing carbon dioxide from flue gases and storing it underground • **adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	
	Model answer - Level 3 (8 marks)	
	Technology has repeatedly extended fossil-fuel supplies. Secondary and tertiary recovery raise the proportion of oil recovered from about 20% to 60%; directional drilling, subsea wells and ROVs open deep-water fields; fracking releases tight oil and gas; tar sands and oil shales, coal gasification and liquefaction convert lower-grade resources; and carbon capture and storage could reduce the CO_2 impact. Nuclear fission can be stretched even further: uranium can be adsorbed from seawater with polymers or recovered from phosphate mining and coal ash, molten-salt reactors run more safely, and plutonium breeder and thorium reactors multiply the energy obtainable from each tonne of fuel, with fusion a long-term prospect.	
	None of this makes the resources renewable. Discoveries of conventional oil have declined since the 1980s, unconventional sources cost more energy and money per unit and damage aquifers, use huge volumes of water, cause tremors and take land, and CCS is unproven at scale. Nuclear projects overrun their budgets, decommissioning and long-lived waste remain unsolved, public opposition and proliferation constrain expansion, low-grade uranium has a high embodied energy, and fusion is not commercial.	
	Sustainability means maintaining supply without unacceptable environmental, economic or social impact. New technology can extend supplies for decades and cut some impacts, which matters for closing the energy gap while renewables and conservation scale up, but it cannot make non-renewable resources sustainable indefinitely. Their proper role is as a transition and base-load source alongside renewables, not a permanent solution.	
	Weaker answer - Level 2 (5 marks)	
	New technologies mean fossil fuels can last longer. Fracking gets gas out of shale rock and directional drilling reaches oil under the sea. Carbon capture and storage can stop the carbon dioxide going into the atmosphere. For nuclear power, breeder reactors and thorium reactors can use fuel more efficiently and fusion might work in the future. However fossil fuels are still non-renewable so they will run out eventually, and fracking can pollute groundwater. Nuclear waste is still a problem. So new technologies help but they cannot make fossil fuels and nuclear power fully sustainable.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 2 reaches a sensible conclusion but supports it thinly; Level 3 gives specific evidence (20% to 60% recovery, declining discoveries, embodied energy of low-grade uranium).	
	• Level 3 sets out a criterion for sustainability and judges against it - the hallmark of 'evaluate'.	
	• Level 3 places the technologies in context (transition and base load alongside renewables) rather than treating them in isolation.	
10.2	1 mark for the point below. **1** 80-90% **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 183-196; AQA spec 3.3.3-3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Fossil fuel features that shaped society: energy density, storage, transport, high temperatures, liquid vehicle fuels, established infrastructure • **Renewable features:** intermittent / unpredictable (wind, solar) vs predictable (tidal) vs controllable (HEP, biofuel, geothermal) – low energy density – location – low pollution but manufacturing impacts • Technologies and developments for each resource with examples – new technologies improving usability • **Storage and grid management:** pumped storage, batteries, hydrogen economy, P2G, V2G, thermal storage, peak shaving – the role of electricity as a secondary fuel • **Economics and politics:** development costs, interest, subsidies, external costs of fossil fuels, falling renewable costs • Energy conservation as part of the solution – sustainability principles - natural systems run on low-density renewable energy • **Judgement:** electricity largely yes	25
	– heat, aviation, shipping and high-temperature industry harder – transition period with nuclear / CCS *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Stages of investigation:** identifying the issue (Halley Station UV data, Dobson unit measurements), hypothesis formation (Rowland-Molina 1974: persistence, UV dissociation, chlorine reactions), data collection (ground, satellite, balloon and aircraft air sampling detecting ClO), analysis of spatial, temporal and altitude variability, proposal of solutions, enactment • **Why Antarctica:** low stratospheric temperatures, ice crystals, stratospheric clouds, polar vortex, spring sunlight • **Effects of UVB:** skin cancer, cataracts, DNA damage, plant and marine damage - motivating action • **Solutions:** Vienna Convention, Montreal Protocol - phase-out and bans, alternative processes (pump sprays), alternative materials (HCFCs, HFCs, hydrocarbons, HFAs), collection and incineration of CFCs, fund for developing countries • **Evaluation of success:** ozone hole area / depth stabilising, CFC levels falling	25
	– limitations - persistence, remaining ODSs, HFCs as greenhouse gases, decades to full recovery • Comparison with other atmospheric problems (climate change, acid rain) - why ozone was easier – role of peer review, international cooperation, availability of substitutes • Conclusion on the extent to which it is a model for other issues *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113; AQA spec 3.2.1.3.	

Do not write outside the box